

SELECTED ROBOTICS AI & SYSTEMS INTEGRATION PORTFOLIO

Brandon Gerbasi

“The part I enjoy most is final integration - taking AI, software, sensors, compute, electronics, and hardware and bringing the complete machine to life.”

Computer Science B.S. expected 2026 | Mathematics B.S. in progress | Relocating to Austin, TX

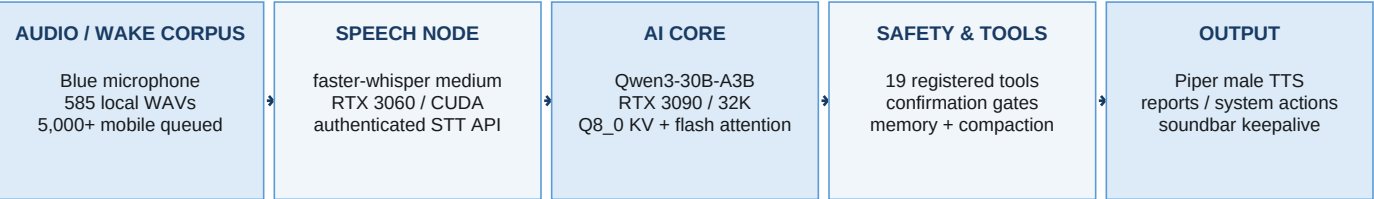
AI & Learning Qwen3 local LLMs, PyTorch, speech AI, wake-word training, RL/IL foundations	Robotics Software C++17, Python, ROS2, state estimation, telemetry, safety logic	Physical Integration Sensors, Jetson/NVIDIA GPUs, UAV hardware, CAD, wiring, fabrication
Systems & Deployment Linux, CUDA, systemd, authenticated APIs, multi-node health, testing	Field Experience Tesla manufacturing, customer-facing field work, travel, rapid cross-training	Current Direction Physical AI, robotics systems integration, edge autonomy, sim-to-real

2 Linux GPU nodes	30B local MoE brain	32K validated context	19 registered tools	15 automated tests
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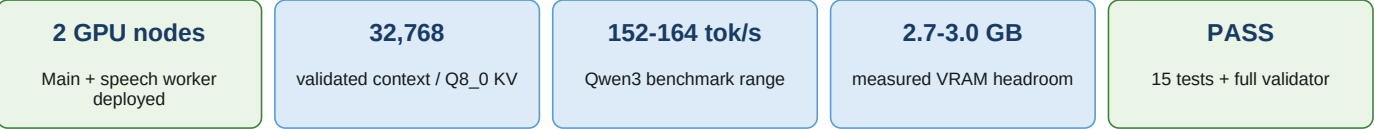
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PROJECT 1 - OTTO: DISTRIBUTED LOCAL AI & ROBOTICS ASSISTANT

Purpose: Build a private, offline-first assistant that can hear, reason, use tools, coordinate across compute nodes, and safely interact with computers and future robotic systems without cloud inference.



VERIFIED BUILD STATUS



WHAT I BUILT

- Deployed a two-node local architecture with Qwen3-30B-A3B on RTX 3090 Main and faster-whisper medium on RTX 3060 Node 2.
- Built an authenticated raw-WAV STT service with **/health**, request limits, CUDA inference serialization, and systemd autostart.
- Refactored the original monolith into modular LLM, STT, TTS, memory, prompt, tool, system, and operations services with a 19-tool registry.
- Implemented protected confirmations, remote service restart, verified Node 2 reboot/recovery, bounded tool loops, atomic state, and history compaction.
- Optimized Qwen3 with the embedded chat template, full GPU offload, Q8_0 KV cache, flash attention, and automatic 14B fallback profiles.
- Loaded Piper once, chunked long speech, and used a persistent audio keepalive to eliminate clipped sentence beginnings and endings.

NEXT ENGINEERING MILESTONES

Strict Offline Lockdown Local model paths, public-network tool controls, and WAN-independent validation.	Local Knowledge & Memory Approved-folder indexing, structured memory, tasks, and project retrieval.
Production Wake Word Import 5,000+ mobile WAVs, deduplicate, normalize, train, and measure false activations.	Custom Voice & Robotics Train Otto's voice and connect real drone/smart-home protocols behind safety interlocks.

PROJECT 2 - CUSTOM FIELD ROBOTICS UAV PLATFORM

Purpose: Design and integrate a custom aerial robotics platform combining propulsion, flight control, telemetry, sensing, edge compute, safety logic, and staged autonomy/vision capabilities.

Mechanical Platform	Custom CAD and 3D-printed body/wings, carbon-fiber reinforcement, side-by-side battery bay, payload and balance planning
Power & Propulsion	Dual CNHL 9500mAh 6S architecture, Tekko32 65A ESC, 2807 1300KV motors, XT90 anti-spark, power-module and protection planning
Flight & Communications	Flight Core V2, u-blox M10 GPS, ELRS 915 MHz control, 915 MHz telemetry, mission/log workflow
Perception Sensors	RP-LiDAR S2, TFmini Plus forward/downward ranging, FLIR Lepton thermal camera, GoPro/visual payloads
Edge Compute	NVIDIA Jetson Orin Nano for ROS2, OpenCV, tracking, obstacle avoidance, sensor fusion, and autonomy work
Ground / Operator Tools	Python telemetry ground station, warnings, reports, logs, state estimation, and repeatable mixed-fault tests

6S x2

battery architecture

2807

1300KV motors

Jetson

Orin Nano edge AI

LiDAR

thermal + range sensing

INTEGRATION STATUS

- Completed initial manual flight/propulsion validation; the custom airframe and wings are being revised and reprinted for repeatable structured testing.
- Core flight-control, power, GPS, control-link, and telemetry architecture is integrated; Jetson and perception hardware are staged for activation.
- Next phase: stable-airframe flight logs, telemetry review, then staged ROS2, computer vision, object tracking, avoidance, and autonomous behaviors.
- The project is intentionally end-to-end: design, fabrication, wiring, software, sensing, testing, failure analysis, and operator tools belong to one system.

WHY THIS PROJECT MATTERS FOR INTEGRATION ROLES

It demonstrates the work I want to own professionally: integrating sensors, compute, control software, electronics, communications, mechanical hardware, and safety into a machine that must work outside a perfect lab environment.

SUPPORTING ROBOTICS & AI PROJECTS

C++ Robot State Estimator	C++17, CMake, GTest	Tracks position, heading, and velocity; handles GPS loss; scores confidence; produces repeatable CSV/Markdown/JSON reports and fault-test artifacts.
UAV Telemetry Ground Station	Python, tests, log analysis	Simulates/replays flight states, detects unsafe or high-current conditions, and generates operator-facing reports across normal and mixed-fault scenarios.
ROS2 Perception & Safety Demo	ROS2 Jazzy, C++	Separates range sensing, IMU input, safety control, and logging; demonstrates CLEAR, SLOW_DOWN, STOP_OBSTACLE, and STOP_TILT states.
Vision-Based Object Tracker	Python, OpenCV	Demonstrates real-time visual tracking and establishes the perception foundation for future UAV/robot autonomy.
Edge-Deploy Safety Monitor	Python, automated tests	Evaluates sensor/system conditions and produces explicit safe-state outputs for edge deployment.

VERIFIED SYSTEMS EVIDENCE

19 registered Otto tools	15 automated tests	2 Linux GPU nodes	32K validated context	~153-164 tokens/sec benchmark
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PROFESSIONAL EVIDENCE

- **Tesla Gigafactory Nevada:** four performance shout-outs in the first month; selected for overtime in the first week; cross-trained across multiple production areas earlier than standard.
- **Botco embodied-AI project:** wore sensing equipment while performing repeatable physical tasks to produce human-demonstration training data.
- **Six years of independent field/customer work:** route planning, deadlines, difficult interactions, documentation, and professional work without close supervision.
- **Hands-on technical background:** electronics, soldering, vehicle repair, heavy machinery operation/maintenance, CAD, fabrication, and Linux/GPU systems.

WHAT I BRING TO A ROBOTICS INTEGRATION TEAM

Systems View Trace failures across software, sensors, power, communication, compute, audio, and mechanics instead of treating one layer in isolation.	Rapid Learning Learn unfamiliar systems by building, testing, documenting, and iterating until the cause is understood - not just the workaround.	Physical Ownership Comfortable wiring, mounting, configuring, programming, testing, and repairing complete machines.
AI + Robotics Connect model training and AI software to ROS2, edge compute, sensor data, safety logic, and real hardware.	Field Readiness Comfortable with travel, customer sites, long deployment days, changing conditions, and direct responsibility for outcomes.	Career Direction Become exceptional at bringing AI-powered physical systems from separate components to reliable operation.

Available to relocate to Austin, TX | Available for field travel | github.com/gerbasi117